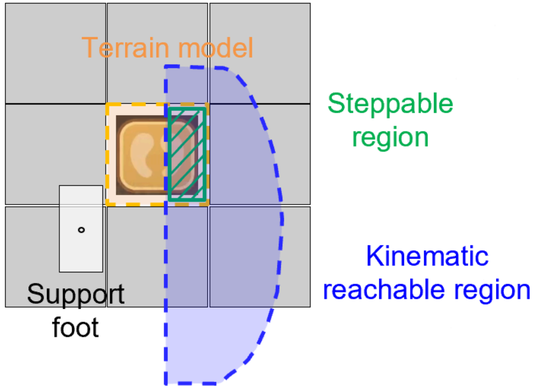
Step 1: Formulate a nonconvex QCQP for terrain feasibility and reachability

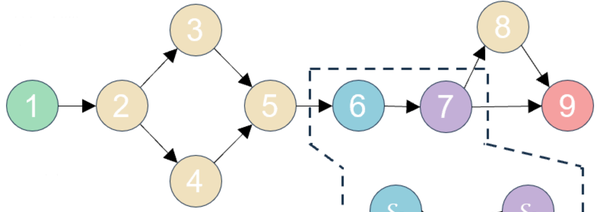
Bilinear couplings and quadratic equality constraints make the footstep relation model nonconvex.



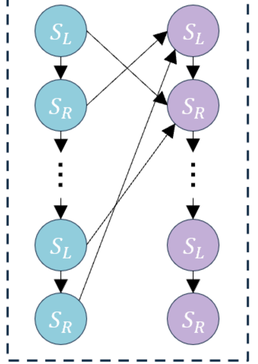
The problem integrates terrain constraints, reachability, and cost in a single optimization problem.

Step 2: Building a graph





Each node encodes a support mode (D, S_L, S_R) and a convex terrain region. Edges encode feasible transitions and allowable gait-mode switches.



Step 3: Solve with the Graph of Convex Sets

(i) Through **Steps 1** and **2**, we cast footstep planning as a shortest-path problem in a Graph of Convex Sets (SPP in GCS).

⇒ *Mixed Integer Nonconvex Program (MINCP)*

↓

(ii) Using Semidefinite programming (SDP) relaxation, it becomes Mixed Integer Convex Program (MICP) with a strong convex relaxation.

⇒ *Mixed Integer Convex Program (MICP)*

↓

(iii) Solve the relaxation, recover a discrete path via relaxation-guided randomized search, and solve the fixed-path convex program for the final footstep plan

⇒ *Relaxed - Mixed Integer Convex Program (R - MICP)*